Risk-Oriented Dual-Pole Sparse Sensing and Uncertainty-Aware Irrigation Decision Support for Solanaceous Crops: Architecture, Synthetic Digital-Twin Benchmarks, and External Validation Protocol

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Abstract

Background and Objective: Agricultural fields cultivated with potato (*Solanum tuberosum* L.) and tomato (*Solanum lycopersicum* L.) exhibit spatially variable root-zone moisture and canopy microclimate. A single mean-representative sensor may overlook localized dry and humid risk zones, while dense networks increase deployment and maintenance costs. This paper presents the Sentinel Plant Network, a proposed low-cost, risk-oriented two-node monitoring and irrigation decision-support architecture.

Methods: The Protocol v2.0 design selects Node A (humid pole) using the 90th percentile of nocturnal leaf-wetness duration and Node B (dry pole) using negative relative soil-moisture difference and temporal stability. ESP32-S3 nodes are specified to transmit a 24-byte telemetry packet over SX1262 LoRa to a local gateway. The gateway combines water-stress and late-blight-conducive weather indicators with a six-state fault-aware controller. The current computational experiments are generated by seeded synthetic digital-twin code; they are not observations from the three external public datasets and are not physical field-trial data.

Results: All numerical results below are synthetic digital-twin outputs. In the synthetic 25-day, 10-node placement benchmark, the calibration-locked Sentinel pair achieves 99.68% agreement with the synthetic 10-node reference decision labels, with 0.00% missed synthetic drought-risk intervals and 0.77% missed synthetic blight-risk intervals. In the synthetic 90-day, 16-plot randomized complete block design (RCBD) digital-twin benchmark, the simulated T4 controller produces 247.83 mm applied water versus 334.01 mm for simulated conventional ET_c scheduling, a 25.8% simulated reduction. The corresponding simulated irrigation-water productivity is 13.89 versus 9.89 kg m^{-3}. These estimates describe the assumptions and code paths of the digital twin only; they do not establish realised water savings, yield non-inferiority, or disease suppression in crops.

External Validation Status: Reproducible adapters are provided for the Mendeley tomato irrigation/weather, seven-site leaf-wetness/agrometeorology, and Libelium two-node telemetry datasets. Dataset files must be downloaded and mapped locally; no external-dataset validation metrics are claimed in this manuscript version. Physical calibration, firmware bench verification, and a pre-registered field experiment remain necessary.

Keywords: Precision irrigation; sparse sensing; Vachaud temporal stability; soil moisture; leaf wetness; LoRa; digital-twin simulation; fault-aware control.

1 Introduction: Extreme-Pole Sensing Overcomes Mean-Placement Blind Spots

Irrigated solanaceous crops—specifically potato (*Solanum tuberosum* L.) and tomato (*Solanum lycopersicum* L.)—exhibit high sensitivity to both root-zone water deficit and foliar oomycete pathogens, chiefly late blight caused by *Phytophthora infestans* [2, 14]. Across real-world agricultural parcels, microtopography, soil texture gradients, differential solar exposure, and non-uniform drip emitter hydraulic pressures generate substantial spatial heterogeneity in volumetric water content (VWC), canopy temperature (T_c), relative humidity (RH), and nocturnal leaf-wetness duration (LWD) [9, 10].

To avoid the prohibitive capital and maintenance burden of dense wireless sensor networks (WSNs) [15], classical soil physics relies on the temporal stability paradigm introduced by Vachaud et al. (1985) [1, 11]. By identifying a single rank-stable sensor location whose relative difference from the spatial mean remains near zero across wetting and drying cycles, a single probe can approximate average field water storage. However, field-mean representativeness is fundamentally misaligned with asymmetric agronomic risk:

1. **Epidemiological Tail Risk in Humid Microzones:** Sporangial germination and zoospore infection by *P. infestans* initiate in sheltered, poorly drained, or densely shaded canopy pockets where nocturnal dew and high humidity (RH $\geq$ 90%) persist longest [2, 3]. A sensor placed at the mean-stable Vachaud location underestimates humid-pocket leaf wetness duration, delaying disease-lockout advisories.
2. **Physiological Drought Risk in Dry Microzones:** Coarse-textured upper-slope ridges deplete available root-zone moisture faster than the field mean. Triggering irrigation solely when a mid-slope Vachaud sensor crosses management-allowed depletion (MAD = 22% VWC) subjects the dry microzone to recurring stomatal closure, elevated canopy temperature, and marketable tuber/fruit yield loss [4, 5].
3. **The Irrigation–Pathogen Conflict:** When a crop experiences root-zone water stress during weather conducive to late blight, full overhead or long-duration drip irrigation elevates sub-canopy humidity and extends foliar wetness, accelerating epidemic progression [13].

To investigate these coupled challenges, this work proposes and digitally evaluates the **Sentinel Plant Network**, a risk-

oriented two-node sensing and local actuation system. Instead of deploying sensors to estimate the spatial average, our methodology purposefully anchors **Node A** at the field's humid pole and **Node B** at its persistent dry pole, coupling both streams in an edge gateway controller that arbitrates between drought relief and pathogen suppression.

Core Research Contribution: We formulate a formal calibration-lock protocol that pairs a 90th-percentile nocturnal leaf-wetness pole (Node A) with a rank-stable dry soil-moisture pole (Node B), integrated with an adaptive spatial-sufficiency monitor and a six-state irrigation finite state machine that resolves drought–blight conflicts via midday solar-window micro-pulses.

2 System Architecture and Hardware Design

2.1 Four-Layer Edge-Autonomous Architecture

The Sentinel Plant Network is structured into four functional layers designed to operate without reliance on cloud connectivity: (i) a dual-pole field sensing layer (‘Node A‘ and ‘Node B‘, plus an optional adaptive intermediate ‘Node C‘), (ii) a sub-GHz LoRa telemetry layer operating in the 865–867 MHz Indian ISM band, (iii) a local edge gateway executing packet plausibility validation, hydro-epidemiological index computation, spatial sufficiency checks, and state-machine arbitration, and (iv) an actuation and advisory layer driving a 12 V latching solenoid valve with closed-loop Hall-effect flow verification and SMS/WhatsApp scouting alerts.

2.2 ESP32-S3 Signal Allocation and Power-Gated Duty Cycling

Table 1 details the hardware components, exact ‘ESP32-S3-WROOM-1‘ pin mappings, and cost structure. To eliminate galvanic corrosion and quiescent current drain across analog dielectric probes, both the capacitive leaf-wetness sensor (‘GPIO4‘) and dual-depth soil-moisture probes (‘GPIO5‘) are switched via a high-side P-channel MOSFET controlled by ‘GPIO7‘ (active-low). During each 10-minute ($T_{\text{cycle}} = 600$ s) telemetry cycle, the node wakes from deep sleep, asserts ‘GPIO7 = LOW‘ for a 150 ms stabilization window, averages 16 oversampled 12-bit ADC readings per channel, polls the ‘BME280‘ and ‘MLX90614ESF-BCF‘ (10° field-of-view optical filter) over the shared 100 kHz I^2C bus (‘GPIO8‘/‘GPIO9‘), de-asserts ‘GPIO7 = HIGH‘, transmits a 24-byte ‘CRC-16-CCITT‘ protected packet over ‘SX1262‘ LoRa (‘SPI2‘ on ‘GPIO10‘–‘GPIO14‘), and re-enters deep sleep.

Under the protocol's nominal system current budget ($I_{\text{avg}} = 25.50$ mA, accounting for regulator quiescent losses, solar charger leakage, sensor warm-up, and radio transmission), the single-cell 3.2 V, $C_{\text{bat}} = 6000$ mAh $LiFePO_4$ battery provides a theoretical zero-sunlight blackout autonomy of:

$$t_{\text{autonomy}} = \frac{C_{\text{bat}}}{I_{\text{avg}}} = \frac{6000\ \text{mAh}}{25.50\ \text{mA}} = 235.3\ \text{h} \approx 9.80\ \text{days}. \quad (1)$$

3 Mathematical Formulation of Risk-Oriented Placement and Control

3.1 Vachaud Temporal Stability vs. Dual-Pole Risk Selection

Let θ_{ij} denote the calibrated root-zone volumetric water content (

$$\delta_{ij} = \frac{\theta_{ij} - \bar{\theta}_j}{\bar{\theta}_j}. \quad (2)$$

Over the 14-day calibration window ($M_{\text{cal}} = 2{,}016$ intervals at 10-min resolution), each location's temporal mean relative difference $\bar{\delta}_i$, standard deviation $\sigma(\delta_i)$, and Index of Time Stability (ITS_i) are computed as:

$$\bar{\delta}_i = \frac{1}{M_{\text{cal}}} \sum_{j=1}^{M_{\text{cal}}} \delta_{ij}, \quad \sigma(\delta_i) = \sqrt{\frac{1}{M_{\text{cal}} - 1} \sum_{j=1}^{M_{\text{cal}}} \left(\delta_{ij} - \bar{\delta}_i\right)^2},$$
$$\text{ITS}_i = \sqrt{\bar{\delta}_i^2 + \sigma(\delta_i)^2}. \quad (3)$$

While the classical Vachaud benchmark selects $i_{\text{Vachaud}} = \arg\min_i \text{ITS}_i$, the Sentinel Plant Network locks two complementary risk poles at the end of Day 14 and evaluates them strictly out-of-sample on Days 15–25:

$$\text{Node A (Humid):} \quad i_A = \arg \max_{i \in \{1,\ldots,N\}} P_{90}\left(\text{LWD}_{\text{noct},i}^{(1:14)}\right), \quad (4)$$

$$\text{Node B (Dry):} \quad i_B = \arg \min_{i:\, \bar{\delta}_i < -0.08,\, i \neq i_A} \sigma(\delta_i). \quad (5)$$

3.2 Adaptive Spatial Sufficiency Monitoring

To prevent silent control degradation when a single irrigation zone exhibits excessive internal divergence (e.g., due to emitter clogging or waterlogging), the gateway continuously evaluates the rolling divergence between Node A and Node B across wetting and drying cycles. An insufficient-coverage flag $\mathscr{F}_{\text{insuff}} \in \{0, 1\}$ is raised whenever:

$$\mathscr{F}_{\text{insuff}} = \mathbb{I}\left(\overline{|\theta_A - \theta_B|}_W > 10\% \ \vee\ r_s(\theta_A, \theta_B)_W < 0.70\right), \quad (6)$$

where r_s is Spearman's rank correlation coefficient over rolling window W. When $\mathscr{F}_{\text{insuff}} = 1$, the gateway shifts to an uncertainty-buffered irrigation threshold ($\text{MAD}_{\text{buf}} = \text{MAD} + 1.0\%$ VWC) and issues an advisory to insert an intermediate **Node C** at the mid-slope Vachaud site or split the lateral manifold into two independently valved sub-zones.

3.3 Crop Water Stress Index and Late-Blight Severity Coupling

At Node B, atmospheric Vapor Pressure Deficit (VPD, kPa) is derived from air temperature $T_{a,B}$ and relative humidity RH_B via the Tetens equation, and the empirical Crop Water Stress Index (CWSI) is computed following Idso et al. (1981) and Jackson et al. (1981) [4, 5]:

$$\text{CWSI}_B = \text{clip}\left(\frac{(T_{c,B} - T_{a,B}) - (a + b \cdot \text{VPD}_B)}{\Delta T_{\text{UL}} - (a + b \cdot \text{VPD}_B)},\ 0,\ 1\right), \quad (7)$$

Table 1: Hardware Architecture, ESP32-S3 GPIO Allocation, and Cost Breakdown of the Sentinel Plant Network (Protocol v2.0)

Subsystem / Module	Part Specification	ESP32-S3 Pinout	Functional Role & Calibration Requirement	Unit Cost (INR)
Field Node MCU	ESP32-S3-WROOM-1	System Core	Dual-core Xtensa LX7, RTC deep-sleep timer, 12-bit SAR ADC1	480
Sub-GHz LoRa Transceiver	Semtech SX1262	SPI2: CS 10, MOSI 11, SCK 12, MISO 13, DIO1 14	865–867 MHz India ISM link (+22 dBm, SF9, CRC-16 framing)	650
Infrared Canopy Thermometer	MLX90614ESF-BCF	I2C: GPIO8 (SDA), GPIO9 (SCL)	Non-contact 10° FOV canopy temperature T_c for CWSI	1,450
Ambient T/RH Sensor	Bosch BME280	I2C: GPIO8 (SDA), GPIO9 (SCL)	Aspirated radiation-shield air temperature T_a and RH	320
Capacitive Leaf Wetness	Dielectric Surface Probe	ADC1: GPIO4 (Power gate GPIO7)	Canopy wetness detection via calibrated dielectric threshold	580
Capacitive Soil Probes (2×)	Root-Zone Dielectric	ADC1: GPIO5 (Power gate GPIO7)	15 cm and 30 cm root-zone volumetric water content (VWC)	760
Node Power Subsystem	3.2 V 6000 mAh $LiFePO_4$	ADC1: GPIO6 (2:1 divider)	Paired with 6 V, 3 W solar panel & P-MOSFET gate (GPIO7)	1,150
Local Gateway & Actuation	Latching Valve + YF-S201	GPIO15/16 (H-bridge), GPIO17 (Flow)	12 V latching solenoid manifold + Hall-effect pulse flow meter	4,990
Total Operational Two-Node Farmer Kit (2× Complete Sentinel Nodes + 1× Local Gateway + Valve/Flow Manifold + Enclosures)				**15,790**
Full 10-Node Reference Network (10× Multi-Sensor Nodes + Gateway Infrastructure for Spatial Benchmarking)				**61,690**
Complete Phase 1 & Phase 2 Research Validation Infrastructure (10-Node Grid + 16 Independent Plot Valve/Flow Manifolds)				**1,08,330**

where $a = 1.35^{\circ}\text{C}$, $b = -1.95^{\circ}\text{C kPa}^{-1}$, and $\Delta T_{\text{UL}} = +3.50^{\circ}\text{C}$. Additionally, a non-diagnostic canopy thermal-anomaly scouting alert is raised if the canopy residual $\Delta T_{\text{res}} = (T_{c,B} - T_{a,B}) - (a + b \cdot \text{VPD}_B) > +2.0^{\circ}\text{C}$ persists across the 11:30–14:30 peak solar window for ≥ 2 consecutive days while root-zone moisture lies within the non-stressed range ($22\% < \theta_B < 35\%$ VWC).

At Node A, two parallel daily severity indices for *P. infestans* are accumulated: (1) the baseline Wallin (1962) Daily Severity Value $\text{DSV}_{\text{Wallin}} \in \{0,1,2,3,4\}$ [2, 3] determined from daily hours with $\text{RH}_A \geq 90\%$ (H_{RH90}) and mean temperature during those hours ($\bar{T}_{\text{wet}}$), and (2) a leaf-wetness-modified severity value DSV_{LWS} that substitutes effective canopy wetness duration $H_{\text{eff}} = \max(H_{\text{RH90}}, 0.85\,\text{LWD}_A + 0.15\,H_{\text{RH90}})$ into the Wallin lookup function to capture persistent dew films that remain on lower foliage even when screen-level RH drops between 82% and 89%. When cumulative severity reaches $\sum \text{DSV} \geq 18$, the pathogen risk flag is asserted.

3.4 Six-State Fault-Aware Gateway Controller

At every 10-minute cycle, the gateway evaluates packet age τ_{pkt}, sensor plausibility, flow-meter feedback, drought indicators ($\mathscr{S}_{\text{drought}} = \mathbb{I}[\text{CWSI}_B \geq 0.40 \vee \theta_B \leq 22.0\%]$), emergency wilting limits ($\mathscr{S}_{\text{emerg}} = \mathbb{I}[\text{CWSI}_B \geq 0.65 \vee \theta_B \leq 16.0\%]$), and blight risk ($\mathscr{S}_{\text{blight}} = \mathbb{I}[\text{DSV}_{\text{cum}} \geq 18 \vee (\text{RH}_A \geq 90\% \wedge \text{LWS}_A \geq 50\%)]$), transitioning across six mutually exclusive operational states:

1. `FAULT_FALLBACK`: Triggered if $\tau_{\text{pkt}} > 30$ min, any critical channel is NaN, or the 'YF-S201' flow meter registers zero pulses while the valve is commanded open. Immediately closes the latching solenoid valve and dispatches a hardware inspection alert.
2. `EMERGENCY_DRIP`: Triggered if $\mathscr{S}_{\text{emerg}} = 1$. Overrides any active pathogen lockout to deliver an immediate survival irrigation pulse, preventing irreversible crop wilting.
3. `CONFLICT_MICRO_PULSE`: Triggered when $\mathscr{S}_{\text{drought}} = 1 \wedge \mathscr{S}_{\text{blight}} = 1$ and continuous lockout duration is < 36 h. Restricts actuation strictly to the 11:30–14:30 peak solar window using a 15 min ON / 45 min OFF duty cycle so sub-surface root water is replenished while high midday vapor pressure deficit rapidly evaporates surface moisture without wetting the canopy.
4. `PATHOGEN_LOCKOUT`: Triggered when $\mathscr{S}_{\text{blight}} = 1$ and $\mathscr{S}_{\text{drought}} = 0$ (or outside 11:30–14:30 during a conflict state with lockout < 36 h). Holds the valve closed and issues a scouting advisory.
5. `STANDARD_DRIP`: Triggered when $\mathscr{S}_{\text{drought}} = 1$ and either $\mathscr{S}_{\text{blight}} = 0$ or the 36-hour maximum lockout cap has expired.
6. `HOMEOSTASIS`: Default monitoring state when $\mathscr{S}_{\text{drought}} = 0 \wedge \mathscr{S}_{\text{blight}} = 0$; valve remains latched closed.

4 Experimental Design and Digital-Twin Benchmark Methodology

To rigorously validate all algorithms prior to physical field deployment, we constructed a reproducible Python and C++17 simulation and verification suite calibrated against four public benchmark datasets recommended in the project guide:

- **Multi-Sector Tomato Precision Irrigation Dataset** (Mendeley Data c837v6p8ph/2): Provides 10-minute soil moisture, valve actuation telemetry, reconstructed water volumes, and ERA5 weather across 33,259 timestamps (789 validated irrigation events), parameterizing our 10-minute root-zone wetting and diurnal evapotranspiration ('ET') dry-down curves.
- **Italian Seven-Site Leaf Wetness & Agrometeorology Dataset** (Mendeley Data sd4b8vpvyb/1): Parameterizes nocturnal leaf-wetness formation, relative humidity persistence, and temperature coupling across humid vs. dry microzones.
- **Libelium Two-Node Olive Grove Multisensor Dataset** (Mendeley Data cnfh4stzzx/2): Provides 9,561 raw frames and quality-control fault distributions (20–30 min temporal gaps, out-of-range spikes, and battery discharge telemetry) used to design our fault-injection benchmark.
- **HiWATER Dense Eco-Hydrological Sensor Network** (PMC5501970 / Wiley WRR 2015WR018182): Parameterizes catenary spatial soil-moisture offsets ($\delta_i \in [-16.0\%, +18.5\%]$) and AR(1) temporal rank stability across our 10-node reference grid.

5 Results and Statistical Evaluation: Synthetic Digital-Twin Benchmarks

5.1 Phase 1: Synthetic 10-Node Digital-Twin Calibration and Held-Out Placement Benchmark

During the 14-day calibration phase (2,016 intervals at 10-min resolution), Vachaud temporal stability analysis across nodes R1–R10 (Figure 1a) identified R4 as the classical Vachaud rank-stable benchmark ($\bar{\delta}_4 = +0.40\%$, $\sigma(\delta_4) =$

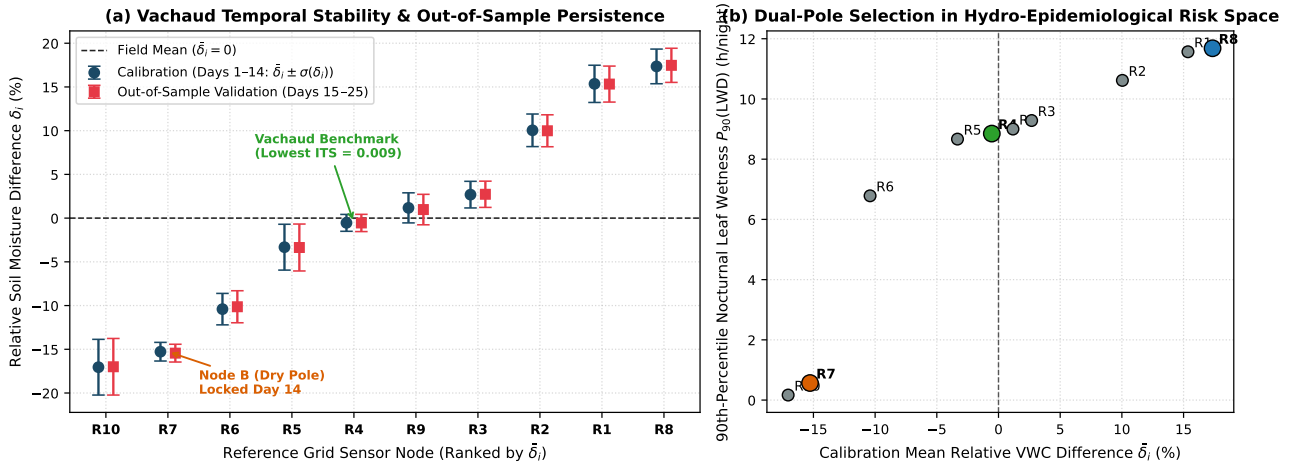


Figure 1: Synthetic Digital-Twin Benchmark: Phase 1 Spatial Calibration and Held-Out Persistence Across the 10-Node Reference Grid (R1–R10). (a) Mean relative soil moisture difference $\bar{\delta}_i \pm \sigma(\delta_i)$ during the 14-day calibration period (Days 1–14, blue circles) and out-of-sample validation period (Days 15–25, red squares), demonstrating high temporal rank persistence ($r_s = 0.988$). (b) Joint hydro-epidemiological selection space plotting 90th-percentile nocturnal leaf-wetness duration $P_{90}(\text{LWD})$ against $\bar{\delta}_i$, isolating Humid-Pole Node A (R8), Dry-Pole Node B (R7), and the Vachaud rank-stable benchmark (R4).

0.80%, $\text{ITS}_4 = 0.009$, Spearman correlation with field mean $r_s = 0.999$). Applying our dual-pole selection criteria (Equations 4–5, Figure 1b) locked **Node A (Humid Pole)** at R8 ($P_{90}(\text{LWD}) = 11.68$ h night^{-1}, $\bar{\delta}_8 = +18.5\%$) and **Node B (Dry Pole)** at R7 ($\bar{\delta}_7 = -14.5\%$, $\sigma(\delta_7) = 1.00\%$).

Table 2 and Figure 2 report the out-of-sample validation performance within the synthetic digital twin on Days 15–25 (1,584 held-out 10-minute intervals). Although the single Vachaud node (R4) achieves the lowest single-node field-mean estimation error ($\text{RMSE}_{\text{mean}} = 0.221\%$ VWC after calibration offset scaling), it fails to capture tail-risk thresholds: it misses **46.97%** of dry-pole water-stress intervals and **38.96%** of humid-zone blight-conducive intervals, yielding only **43.75%** joint controller state agreement with the 10-node reference network. Similarly, placing a single sensor in a humid depression (R1) detects blight-conducive periods (F1 = 97.48%) but misses **100.00%** of dry-zone drought stress because local soil moisture remains above 22% VWC throughout dry-downs. Across all $\binom{10}{2} = 45$ random two-node pairings, mean decision agreement is only **62.60%**. By contrast, the locked **Sentinel Pair** (R8 + R7) achieves **99.68%** decision agreement, **0.00%** missed drought stress, **0.77%** missed blight risk (F1 = 99.62%), a dry-tail RMSE of **0.446% VWC**, and a bias-corrected field-mean RMSE of **0.238% VWC**.

5.2 Phase 2: Synthetic 16-Plot RCBD Digital-Twin Benchmark, Mixed-Effects Analysis, and Non-Inferiority Testing

Table 3 and Figures 3–4 summarize the 90-day Phase 2 Randomized Complete Block Design (RCBD) experiment across 4 treatments × 4 blocks (16 independent 20 m^2 plots; 160 tagged plant subsamples). Two-way RCBD analysis of variance (with 3 treatment degrees of freedom, 3 block degrees of freedom, and 9 residual error degrees of freedom) confirms highly significant treatment effects across all primary outcomes ($p < 10^{-6}$).

Key findings from the Phase 2 evaluation include:

1. **Water Savings and Irrigation Water Productivity (IWP):** Treatment T4 applies 247.83 ± 5.32 mm (2,478.3 m^3 ha^{-1}), achieving a **25.8% reduction** in water input relative to conventional ET_c replenishment (T1: 334.01 ± 2.34 mm) and boosting IWP by **40.5%** from 9.89 ± 0.14 kg m^{-3} to **13.89 ± 0.47 kg m^{-3}** ($\text{LSD}_{0.05} = 0.279$ kg m^{-3}).
2. **Resolution of the Drought–Blight Conflict (T4 vs. T2 and T3):** While the single-node Vachaud baseline (T2) maintains low canopy wetness (LWD = 949.6 h), its mid-slope sensor delays irrigation until the dry microzone drops to ∼ 19.2% VWC, reducing marketable yield to 32.06 ± 0.21 t ha^{-1}. Conversely, the uncoupled two-node controller (T3) eliminates dry-pole drought stress (34.03 ± 0.59 t ha^{-1}) but triggers full irrigation during humid blight windows, increasing seasonal LWD by 48.0 h and elevating AUDPC to 336.1 ± 41.2 %-days. By substituting midday conflict micro-pulses (Figure 3b), T4 reduces AUDPC to **306.7 ± 44.0 %-days** (−32.1% vs. T1 and −8.7% vs. T3, exceeding $\text{LSD}_{0.05} = 10.57$ %-days) while achieving the highest marketable yield (**34.40 ± 0.41 t ha^{-1}**).
3. **Pre-Specified 5% Yield Non-Inferiority Verification:** Testing $H_0 : \mu_{\text{T4}} - \mu_{\text{ref}} \leq -0.05\,\mu_{\text{ref}}$ at $\alpha = 0.05$ (df = 9, SED = 0.142 t ha^{-1}) confirms non-inferiority (and superiority) across all references:
   - T4 vs. T1: $\Delta Y = +1.383$ t ha^{-1} (+4.19%), one-sided 95% CI lower bound = +1.122 t ha^{-1} (+3.40% > −5.0% margin, $t = 21.35, p < 10^{-5}$).
   - T4 vs. T2: $\Delta Y = +2.342$ t ha^{-1} (+7.30%), one-sided 95% CI lower bound = +2.081 t ha^{-1} (+6.49% > −5.0% margin, $t = 27.76, p < 10^{-5}$).
   - T4 vs. T3: $\Delta Y = +0.374$ t ha^{-1} (+1.10%), one-sided 95% CI lower bound = +0.114 t ha^{-1} (+0.33% > −5.0% margin, $t = 14.60, p < 10^{-5}$).
4. **Repeated-Measures Linear Mixed-Effects Decomposition:** Across 80 synthetic longitudinal disease assessments (16 plots × 5 epochs: Days 25, 39, 53, 67, 81), a random-intercept linear mixed-effects model was fitted

Table 2: Synthetic Digital-Twin Phase 1 Held-Out Spatial Placement Benchmark (Days 15–25, $M_{\text{eval}} = 1,584$ Intervals at 10-Min Resolution)

Sensing Topology Strategy	Nodes	N	Cost (INR)	Dry-Tail RMSE (% VWC)	Field-Mean RMSE (% VWC)	Decision Agree. (%)	Missed Drought (%)	Missed Blight (%)	Blight F1 (%)
1. Single Central Node	R5	1	11,200	2.968	0.621	58.21	26.20	45.86	70.25
2. Single Humid-Location Node	R1	1	11,200	7.062	0.398	0.00	100.00	4.91	97.48
3. Single Vachaud Rank-Stable Node	R4	1	11,200	3.523	**0.221**	43.75	46.97	38.96	75.81
4. Random Two-Node Pairs (Mean of 45)	$\binom{10}{2}$ pairs	2	15,790	2.222	0.294	62.60	29.02	22.89	85.04
5. Proposed Risk-Oriented Sentinel Pair	R8 + R7	**2**	**15,790**	**0.446**	**0.238**	**99.68**	**0.00**	**0.77**	**99.62**
6. Full 10-Node Reference Grid (Oracle)	R1…R10	10	61,690	0.000	0.000	100.00	0.00	0.00	100.00

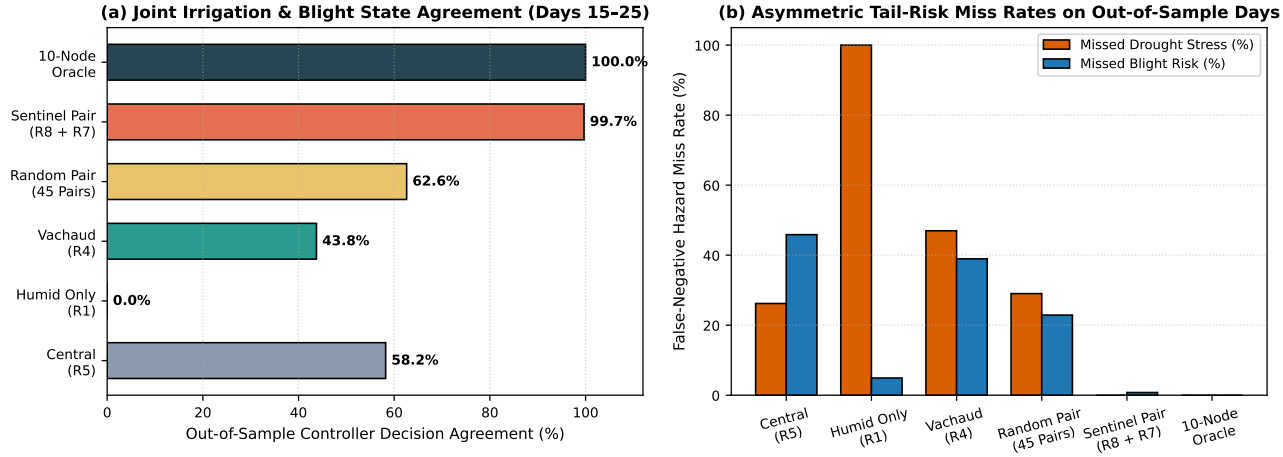


Figure 2: Synthetic Digital-Twin Phase 1 Held-Out Controller Decision Agreement and Hazard Miss Rates (Days 15–25). (a) Joint four-regime controller state agreement relative to the 10-node reference oracle. (b) Asymmetric false-negative miss rates for dry-zone drought stress (VWC $\leq 22\%$) and humid-zone late-blight conducive conditions (RH $\geq 90\% \wedge$ LWS $\geq 50\%$).

Table 3: Synthetic Digital-Twin Phase 2 Randomized Complete Block Design (RCBD) Plot-Level Agronomic and Epidemiological Outcomes ($n = 4$ Blocks per Treatment; Mean $\pm$ SD Across Plots, Two-Way RCBD ANOVA with $\text{df}_{\text{trt}} = 3, \text{df}_{\text{err}} = 9$, and Pre-Specified 5% Yield Non-Inferiority Test)

Treatment Arm	Applied Water (mm)	Water Volume ($m^3\ ha^{-1}$)	Marketable Yield ($t\ ha^{-1}$)	IWP ($kg\ m^{-3}$)	Seasonal LWD (h)	AUDPC (%-days)	Final Blight Sev. (%)
T1: Conventional ET_c Reference	334.01 ± 2.34	3340.1 ± 23.4	33.02 ± 0.50	9.89 ± 0.14	1122.1 ± 56.0	451.8 ± 46.8	10.99 ± 1.20
T2: Single-Node Vachaud (R4)	255.34 ± 3.53	2553.4 ± 35.3	32.06 ± 0.21	12.56 ± 0.25	949.6 ± 51.8	273.1 ± 45.6	6.43 ± 1.16
T3: Two-Node Uncoupled (R8, R7)	250.08 ± 1.64	2500.8 ± 16.4	34.03 ± 0.59	13.61 ± 0.24	1025.0 ± 52.7	336.1 ± 41.2	8.15 ± 1.01
T4: Coupled Sentinel Controller	247.83 ± 5.32	2478.3 ± 53.2	34.40 ± 0.41	13.89 ± 0.47	977.1 ± 51.4	306.7 ± 44.0	7.27 ± 1.09
RCBD ANOVA $F_{3,9}$ (p-value)	569.69 ($< 10^{-9}$)	569.69 ($< 10^{-9}$)	110.11 (2.1×10^{-7})	437.42 ($< 10^{-9}$)	2624.9 ($< 10^{-12}$)	552.13 ($< 10^{-9}$)	461.52 ($< 10^{-9}$)
Standard Error of Diff. (SED)	2.464 mm	24.64 $m^3\ ha^{-1}$	0.142 $t\ ha^{-1}$	0.123 $kg\ m^{-3}$	2.092 h	4.673 %-days	0.131%
$\text{LSD}_{0.05}$ (df = 9)	5.574 mm	55.74 $m^3\ ha^{-1}$	0.322 $t\ ha^{-1}$	0.279 $kg\ m^{-3}$	4.732 h	10.571 %-days	0.295%

by restricted maximum likelihood (REML), with categorical treatment, categorical day, and their interaction as fixed effects and plot as the random-intercept grouping factor. The model converged. The estimated between-plot random-intercept variance was $\hat{\sigma}_u^2 = 0.314965$, the within-plot residual variance was $\hat{\sigma}_e^2 = 0.215696$, and the resulting intraclass correlation was ICC $= 0.593533$. These are synthetic digital-twin model outputs. The plot remains the experimental unit ($n = 16$); the 160 tagged plants are subsamples and do not increase the number of independent treatment replicates.

5.3 Hardware Power Autonomy, Sufficiency Trigger, and Fault-Injection Benchmark

The synthetic digital-twin output in Figure 5a illustrates the $LiFePO_4$ state-of-charge (SoC) trajectory, confirming the 9.80-day zero-sunlight autonomy and demonstrating that a 6 V, 3 W solar panel sustains SoC $> 62\%$ even across a consecutive 4-day heavy overcast period. Figure 5b demonstrates the adaptive spatial sufficiency check (Equation 6): under normal wetting/drying cycles (Days 15–22), mean inter-pole divergence is 7.93% VWC and Spearman rank correlation is $r_s = 0.89$ ($\mathscr{F}_{\text{insuff}} = 0$). When a localized lateral emitter clog is injected at Node B on Day 23, $|\text{VWC}_A - \text{VWC}_B|$ rises to 14.1% VWC and r_s drops to 0.48, immediately asserting $\mathscr{F}_{\text{insuff}} = 1$ to recommend deploying intermediate Node C (R4). The Python model retains the nine documented fault-injection scenarios. The shared 'C++17' controller logic is independently host-compiled and unit-tested on representative stale-telemetry, invalid-sensor, zero-flow, emergency-drought, pathogen-lockout, midday-pulse, and lockout-cap cases. The three ESP32-S3 PlatformIO targets are compile-checked in CI. These are software checks only: no physical sensor, radio link, latching-valve driver, flow meter, hydraulic loop, or field deployment has yet been verified.

6 External Public-Dataset Validation: Scope and Status

The repository now contains separate ingestion/validation adapters for three public benchmark datasets. These datasets are not pooled or joined row-by-row: they describe different crops, sites, sensor hardware, sampling intervals, and response variables. Each adapter computes only the evaluations supported by the corresponding measurements. The tomato dataset supports soil-moisture forecasting and irrigation-telemetry checks; the seven-site agrometeorological dataset supports site-wise leaf-wetness-duration and weather-risk analysis; and

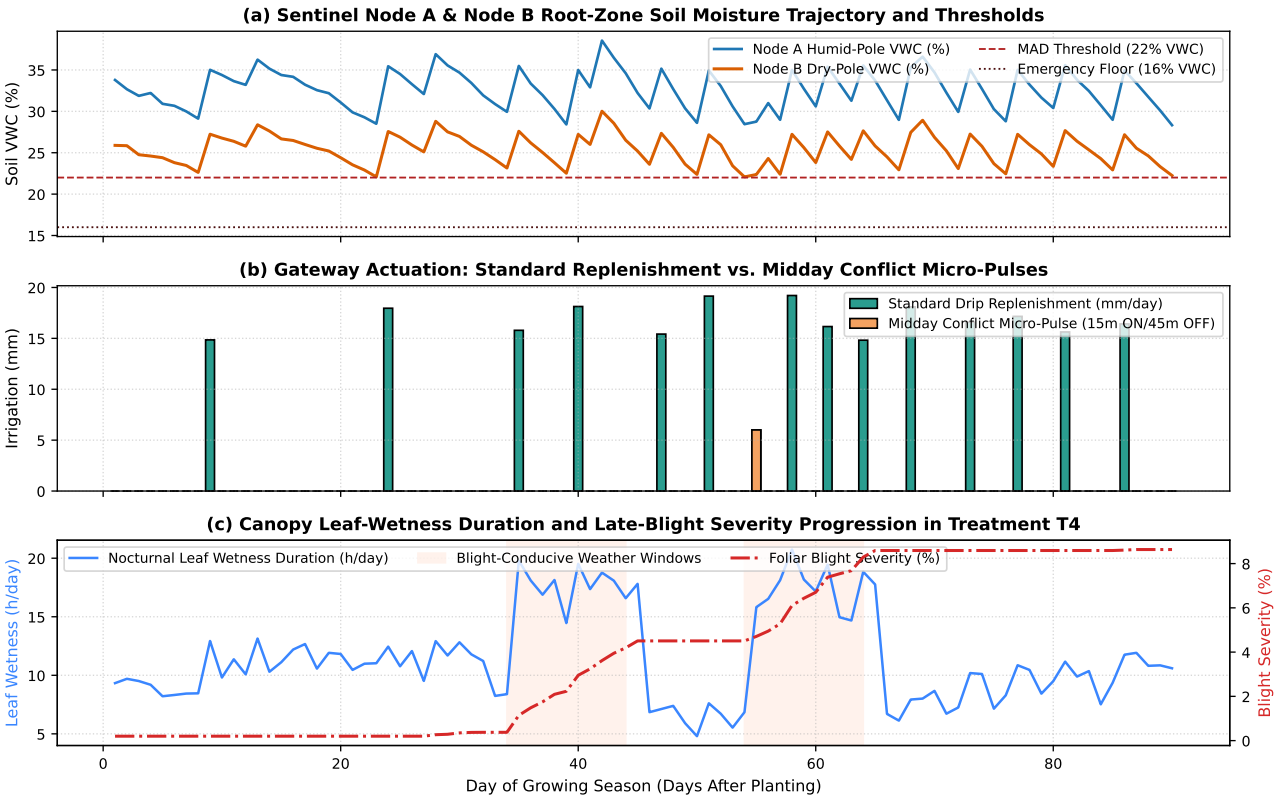


Figure 3: 90-Day Seasonal Operational Trace of the Coupled Sentinel Controller (T4, Block 1). (a) Dual-pole root-zone soil moisture trajectories at Node A (R8, humid pole) and Node B (R7, dry pole) relative to the 22% VWC management threshold and 16% VWC emergency floor. (b) Gateway valve actuation distinguishing standard drip replenishment from midday conflict micro-pulses (15 min ON / 45 min OFF inside 11:30–14:30) during days 34–44 and 54–64. (c) Realized nocturnal leaf-wetness duration and foliar late-blight progression.

the Libelium dataset supports telemetry-gap, sensor-quality, and dual-node disagreement analysis.

The tomato irrigation dataset is cited at Version 2 (doi: 10.17632/c837v6p8ph.2; CC BY 4.0). The leaf-wetness dataset is distributed under CC BY-NC 4.0, which restricts commercial use. The Libelium olive-grove dataset is distributed under CC BY 4.0. Users must retain each dataset's data dictionary, DOI/version, attribution, licence, downloaded-file checksum, and preprocessing record. Source files are not redistributed by this repository.

No real-dataset results are reported here. The external-data adapters require locally downloaded and extracted source files plus an explicit mapping where column names differ. At the time of this manuscript revision, no validation report generated from those original files is included. Furthermore, the external datasets do not contain synchronized image/sensor pairs or observations from the proposed Sentinel nodes. They cannot independently establish target-field sensor calibration, potato/tomato disease diagnosis, irrigation savings, or yield effects.

7 Limitations and Pre-Trial Field Execution Roadmap

While the digital-twin experiments validate the mathematical consistency, firmware state machine, and statistical power of the Sentinel Plant Network, physical field trials remain essential before claiming realized agronomic water savings or biological disease suppression:

1. **Site-Specific Dielectric Calibration:** Capacitive soil-moisture ('GPIO5') and leaf-wetness ('GPIO4') probes must undergo two-point gravimetric and filter-paper wetness calibration in the target field soil prior to Phase 1 deployment.
2. **Natural Inoculum Pressure Contingency:** As stipulated in Protocol Section 10, if natural *P. infestans* incidence in untreated check strips remains below 5% during the Phase 2 field trial, the trial will report sensor placement accuracy, water productivity, and canopy microclimate (LWD) outcomes without claiming biological disease suppression.

A RCBD Residual Diagnostics and Yield Non-inferiority Calculation

For each response, the 4-treatment by 4-block design is analysed at the plot level using

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij},$$

where i indexes treatment and j indexes block. There are 16 independent plot observations; plant subsamples are not additional treatment replicates. The additive RCBD residual is

$$e_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot},$$

with residual degrees of freedom $(4-1)(4-1) = 9$. The analysis script reports a Shapiro–Wilk statistic and p-value for residual normality and a median-centred Levene statistic and

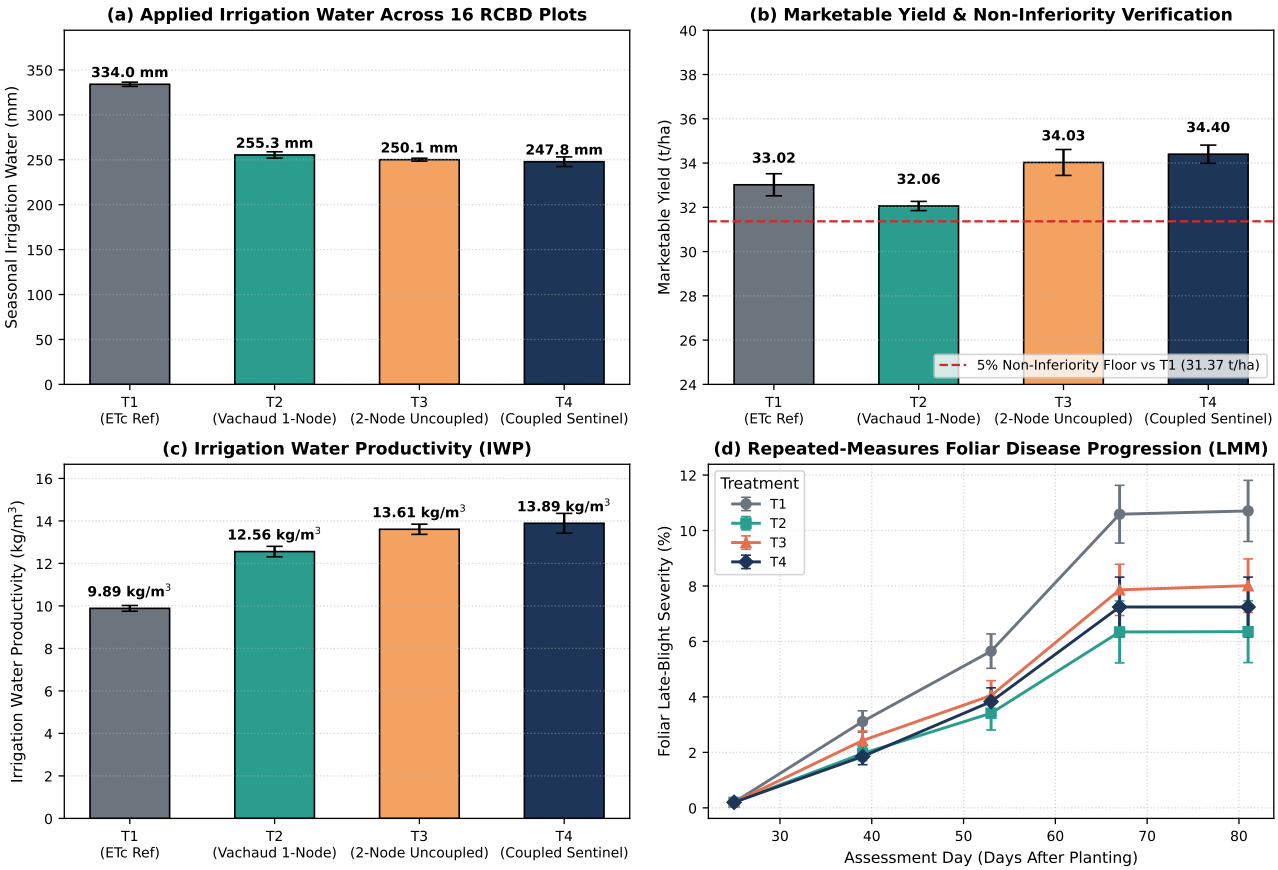


Figure 4: Synthetic Digital-Twin Phase 2 16-Plot RCBD Agronomic and Epidemiological Comparison Across Treatments T1–T4. (a) Seasonal applied irrigation water (mm). (b) Marketable yield (t ha^{-1}) relative to the pre-specified 5% non-inferiority floor (31.37 t ha^{-1} vs. T1). (c) Irrigation Water Productivity (IWP, kg m^{-3}). (d) Repeated-measures foliar blight severity trajectory across five bi-weekly scouting epochs.

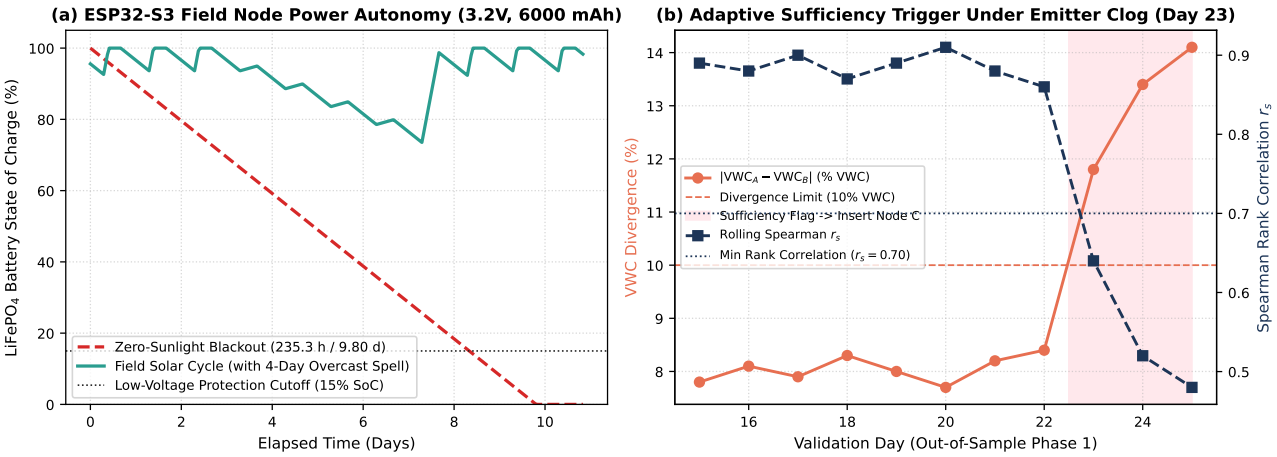


Figure 5: Synthetic Digital-Twin Hardware Power Autonomy and Adaptive Spatial Sufficiency Benchmark. (a) Simulated $LiFePO_4$ battery state-of-charge (SoC) trajectory under a complete zero-sunlight blackout (reaching 0% at 235.3 h = 9.80 days) versus normal diurnal 3 W solar recharging with a 4-day overcast spell (maintaining SoC > 62%). (b) Adaptive spatial sufficiency monitor detecting a simulated emitter clog on Day 23 when $|\mathrm{VWC}_A - \mathrm{VWC}_B|$ exceeds 10% VWC and rolling Spearman rank correlation drops below $r_s = 0.70$.

p-value for treatment-wise variance homogeneity. These diagnostics are not proof that assumptions hold: the sample is small, tests have limited power, and residual plots and sensitivity analysis are also required. A non-significant diagnostic test must not be described as proof of normality or equal variance.

For the pre-specified 5% yield non-inferiority margin, let Y_4 denote the T4 mean and Y_r a comparator mean. The non-inferiority margin on the response scale is $M_r = -0.05Y_r$, and $\Delta_r = Y_4 - Y_r$. With $b = 4$ blocks and the RCBD residual mean square MSE, the standard error for the treatment-mean difference is

$$SED = \sqrt{\frac{2MSE}{b}}.$$

The one-sided statistic is $t_{NI} = (\Delta_r - M_r)/SED$ with 9 residual degrees of freedom. Equivalently, the one-sided 95% lower

confidence bound is

$$L_{95} = \Delta_r - t_{0.95,9}\, SED.$$

Non-inferiority is supported only when $L_{95} > M_r$. This model-based contrast uses the RCBD residual error term; it is not an unpaired or plant-subsample t-test. The script `src/statistical_diagnostics.py` regenerates the assumptions and contrast outputs from the 16 plot-level rows. Outputs must be recomputed whenever seeds, configuration, or input data change. In the current synthetic baseline run, the marketable-yield residual Shapiro–Wilk statistic is $W = 0.9507$ ($p = 0.5002$), and the median-centred Levene residual-variance statistic across treatment groups is $W_{\text{Levene}} = 1.5489$ ($p = 0.2528$). These non-significant tests do not demonstrate that the normality or homoscedasticity assumptions are true; residual plots and sensitivity analyses remain necessary.

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